



Residual Toxicity and Eco-toxicological Assessment of Commonly Used Disinfectants in Sericulture

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Abstract

Disinfection forms the cornerstone of disease management in sericulture, where microbial pathogens pose constant threats to healthy silkworm rearing. Although chemical disinfectants such as formalin, bleaching powder, iodine preparations, and quaternary ammonium compounds (QACs) remain indispensable for maintaining hygiene, their residual toxicity and eco-toxicological effects are not sufficiently addressed by farmers or extension systems. Chemical residues may persist on rearing appliances, contaminate soil and water sources, affect non-target organisms, and undermine environmental safety. Silkworms (*Bombyx mori* L.), being highly sensitive to chemical pollutants, are particularly vulnerable to sub-lethal effects, resulting in reduced growth, poor cocoon quality and impaired silk filament characteristics. This expanded review examines the toxicological properties of widely used disinfectants, their residual behavior, impacts on silkworm biology, influence on soil and aquatic ecosystems and associated occupational hazards. This article underscores the urgent need for standardized monitoring protocols, farmer education, and the adoption of safe, eco-friendly disinfection technologies to achieve sustainable and environmentally responsible sericulture.

Keywords: Sericulture, disinfection, residual toxicity, eco-toxicology, formalin, bleaching powder, iodophore disinfectants, quaternary ammonium compounds, silkworm health, cocoon quality, environmental contamination, soil microbes and herbal disinfectants.

Introduction

Sericulture is a microbe-sensitive bio-industry where even minor lapses in sanitation may lead to devastating crop losses. Pathogens such as *Nosema bombycis* (pebrine), viruses causing nuclear polyhedrosis, bacteria, fungi, and other contaminants thrive in poorly sanitized rearing houses. Thus, chemical disinfection has traditionally been an integral part of pre-rearing and inter-batch cleaning practices.

Commonly used disinfectants-formalin, bleaching powder, iodine solutions, and QACs-have strong antimicrobial efficacy but pose significant risks when applied without proper knowledge of recommended concentrations, exposure duration and aeration requirements. As sericulture shifts toward semi-commercial and commercial models, understanding the environmental behavior, residual toxicity and ecological footprint of these disinfectants becomes crucial. Moreover, excessive dependence on harsh chemical disinfectants contradicts principles of eco-sericulture and environmental sustainability.

This expanded review synthesizes existing literature to highlight gaps, risks and sustainable alternatives for safer disinfection practices.

Common Disinfectants Used in Sericulture

Overview of Major Disinfectant Groups

Disinfectant	Active Ingredient	Mode of Action	Typical Use in Sericulture
Formalin (37–40% formaldehyde)	Formaldehyde gas in aqueous solution	Protein coagulation, nucleic acid denaturation	Fumigation of rearing rooms and appliances
Bleaching Powder (Calcium hypochlorite)	Chlorine compounds	Oxidation of cellular components	Floor cleaning, utensil disinfection
Iodine/Iodophore preparations	Iodine complexed with surfactants	Iodination of proteins and enzymes	Disinfection of trays, eggs, and rearing equipment
Quaternary Ammonium Compounds (QACs)	Benzalkonium chloride and related molecules	Disruption of cell membranes	Surface disinfection, equipment sanitation
Herbal & Bio-based Disinfectants	Neem, turmeric, tulsi extracts and phytochemicals	Antimicrobial action via phenolics, terpenoids	Eco-friendly rearing room sanitation

These disinfectants vary in antimicrobial strength, persistence, environmental behavior, and safety levels. While formalin and chlorine compounds provide high efficacy, their toxicity and persistence raise environmental concerns. Herbal disinfectants and probiotic-based cleaners are gaining attention as safer alternatives.

Residual Toxicity in Sericulture

Impact on Silkworm Physiology and Performance

Silkworms are extremely sensitive to chemical contaminants due to their delicate integument and tracheal system.

Effects of Formalin Residues

- Persistent vapors may remain adsorbed on wall surfaces, appliances, and flooring.
- Exposure leads to:
 - Reduced feeding and sluggishness
 - Extended larval duration
 - Increased mortality during late instars
 - Malformed cocoons or thin-shelled cocoons
- Sub-lethal exposure affects silk gland development, reducing filament length.

Effects of Bleaching Powder (Chlorine Compounds)

- Irritates tracheal openings, causing spasmodic movements and restlessness.
- Oxidative stress affects larval tissues, impairing digestion and nutrient assimilation.
- Excessive residues on leaves reduce larval appetite, leading to lower cocoon weight.

Effects of Iodine-Based Disinfectants

- Comparatively safer, but high concentrations can cause:
 - Oxidative stress
 - Minor burns or discoloration on larvae
 - Impaired spinning activity in extreme cases

Sub-Lethal Effects Documented in Studies

- Decline in cocoon weight by 10-18% due to formalin exposure.
- Reduced filament strength due to chlorine adherence on cocoon surfaces.
- Low reelability and increased breaks during reeling.

Effect on Cocoon and Silk Quality

Chemical residues interfere with:

- Sericin and fibroin integrity, causing weak filament strength.
- Reliability, especially in chlorine-rich environments due to surface hardening of cocoons.
- Color and luster, occasionally leading to slight discoloration.

In commercial silk production, even minor deviations affect market value, demonstrating the economic risk of residual toxicity.

Residual Accumulation in Rearing Environment
Formaldehyde Accumulation

- Repeated fumigation without proper aeration leads to chronic exposure.
- Sub-lethal exposure affects multiple generations, decreasing disease resistance.

Chlorine-based By-products

- Reaction with organic matter results in formation of trihalomethanes (THMs).
- Many THMs, such as chloroform, are carcinogenic and environmentally persistent

Accumulation may degrade overall ecosystem quality in long-running rearing centers.

Eco-toxicological Concerns
Impact on Soil Ecosystems

- Chlorine and formaldehyde reduce microbial biomass and diversity.
- Adversely affects:
 1. Nitrogen-fixing bacteria (*Azotobacter*, *Rhizobium*)
 2. Mycorrhizal fungi
 3. Decomposers responsible for litter degradation
- Long-term use leads to soil acidification and reduced fertility.

Impact on Water Ecosystems

- Disinfectant runoff enters drains, ponds, and irrigation systems.
- Even 0.1 ppm chlorine can impact fish gill activity and amphibian larvae.
- THMs and formaldehyde contribute to:
 - Bioaccumulation
 - Disruption of aquatic food webs

Impact on Non-target Terrestrial Organisms

- Pollinators may be indirectly affected if residues drift to flowering plants.
- QACs are toxic to some beneficial insects and earthworms.

Human Health Risks

Sericulture workers are often exposed repeatedly due to poor ventilation and lack of protective gear.

Short-term exposure

- Eye irritation
- Dermatitis
- Throat inflammation
- Headache or dizziness

Long-term exposure

- Chronic respiratory issues from inhaling formaldehyde vapors.
- Increased cancer risk formaldehyde is classified as Group 1 carcinogen (WHO/IARC).
- QACs can cause chronic dermatitis and respiratory sensitivity.

Mitigation Strategies

Improved Disinfection Practices

- Strict adherence to recommended concentrations (e.g., 2% formalin for fumigation).
- Proper aeration for at least 24-48 hours post-fumigation.
- Avoid mixing different disinfectants without guidance.
- Use PPE such as masks, gloves, and aprons.

Adoption of Eco-friendly Alternatives

Herbal Disinfectants

- Neem, turmeric, tulsi and garlic extracts show broad-spectrum antimicrobial activity.
- Leave minimal residues.
- Safe for workers, silkworms and the environment.

Bio-based Products

- Probiotic cleaners suppress pathogens by competitive exclusion.
- Support healthy microbial communities in the rearing environment.

Physical Disinfection Methods

- UV sterilization
- Steam cleaning
- Heat-based sterilization for trays and appliances

These reduce reliance on chemical disinfectants.

Monitoring and Policy Recommendations

- Periodic testing of air quality, surface residues, and soil/water contamination.
- Training programs for sericulture farmers on:
 - 1.Safe handling
 - 2.First-aid
 - 3.Correct disinfectant preparation
- Development of region-specific regulatory guidelines.
 - 1.Soil microbial health metrics
 - 2.Aquatic toxicity thresholds

Conclusion

Chemical disinfectants remain essential in preventing devastating microbial diseases in sericulture. However, their potential for residual toxicity and ecological damage cannot be ignored. Sustainable sericulture demands a balanced approach-using chemical disinfectant judiciously, ensuring safe practices, adopting eco-friendly alternatives and implementing regular environmental monitoring. Ultimately, improving awareness among farmers and integrating scientific innovations can ensure productivity while safeguarding environmental integrity.

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CATALOGUES